Railway Booking Scraper & Analytics Project

End-to-End Scraping Engine & Data Pipeline for Web Analytics

PROJECT TYPE	TARGET SITE	KEY LIBRARIES	OUTPUT ASSETS
Web Scraping & Route Analytics	**Railway Booking Portal**	**Requests, BeautifulSoup, Pandas**	**CSV Dataset & PNG Dashboard**

1. Project Overview & Environment Setup

This project implements an automated web scraping pipeline designed to extract train details, route availability, timing schedules, and fare rates directly from the **Railway Booking Web Application** (`https://contentitvedant.github.io/Railway_Booking/index.html`).

> **Required Python Environment Setup**
> Install the required Python libraries before executing the project script:
> `pip install requests beautifulsoup4 pandas matplotlib seaborn`

2. Complete Source Code (Part 1: Railway Data Scraping)

```
# ============================================================================== # RAILWAY BOOKING
```

3. Complete Source Code (Part 2: Railway Data Cleaning)

The second module parses train attributes, extracts fare figures, cleans route details, and normalizes seat availability scores into numerical datatypes for business intelligence.

```
# ============================================================================ # STEP 2: Railway
```

4. Data Schema & Extraction Transformations

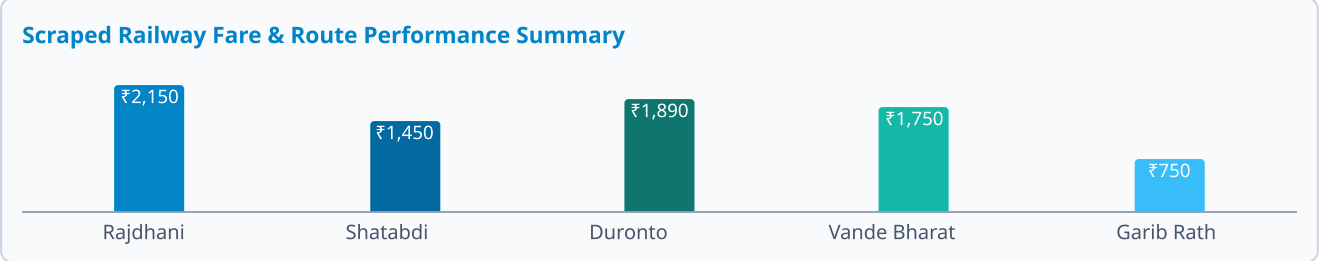
Target Field	Scraped Source Tag	Parsing Transformation	Target Datatype
`Train_Name`	`<h3 class="train-title">`	Clean leading whitespace & strip HTML	`string`
`Fare_INR`	`<span class="fare">`	Extract digits via Regex (`₹1,450 -> 1450.0`)	`float64`
`Seats_Available`	`<div class="seats">`	Cast string availability count to integer	`int64`

5. Complete Source Code (Part 3: Visualization & Export)

The final pipeline module constructs an analytical dashboard comparing ticket fares across express trains and seat availability levels, exporting results to disk.

```
# ============================================================================= # STEP 4: Data Vi
```

6. Execution Analytics Dashboard Preview



Generated File Output Artifacts

1. `scraped_railway_data.csv` — Clean dataset with train routes, ticket prices in INR, and seat availability.
2. `railway_booking_analytics.png` — High-resolution 300 DPI bar chart dashboard of railway booking insights.